

PROFESSIONAL NURSING COACH

QUANTITATIVE APTITUDE

PNC ULTRA-LMR HYBRID MASTER COMPENDIUM • NN-01 • VERIFIED

Arithmetic + commercial + applied + algebra + DI • Batches 1–6 • 72 LMR pages + cover & TOC • NORCET / RRB / ESIC / DSSSB

Batch	Units	Pp
B1	A: numbers, divisibility, HCF/LCM, fractions	12
B2	A-B: fractions, ratio, percent, averages	12
B3	B-C: percent, average, P&L, SI/CI, partnership	12
B4	C-D: mixtures, age, work, pipes, speed, trains	12
B5	D-E: boats, clocks, calendar, algebra, 2D	12
B6	E-F: 3D volumes, melting, DI graphs	12

WhatsApp DM: +91 90571 22020 • Hybrid compact: 2 topics/page, formula + example + trap • v1.0 Sep-2026

Pp	Block	Focus
4-15	A (B1)	Numbers, divisibility, HCF/LCM, fractions
16-27	B (B2)	Fractions, ratio, percent, averages
28-39	C (B3)	P&L;, SI/CI, partnership, mixtures
40-51	D (B4)	Age, work, pipes, speed, trains
52-63	E (B5)	Boats, clocks, calendar, algebra, 2D
64-75	E-F (B6)	3D volumes, melting, DI graphs

How to use: formula first, example second, trap third, speed-check close. Double-QA pending sign-off on worked answers.

Q	Standard	Status
Q1	Sections A-F mapped, batches 1-6	Done
Q2	All worked answers rechecked	Needs sign-off
Q3	Compact 2/page, fonts 10.5-21pt	Done

Natural (1,2,3..) + whole (+0) + integers ($\pm$) + rational (p/q !) + irrational ($\sqrt{2}, \pi$!) — p/q + terminating + closure decides family! Example: $7/3$ rational; $\sqrt{7} + \pi$ irrational.

**N/W/Z + p/q
= rational**

**Irrational: non-end
non-repeat + surds**

**Closure + order
+ audit!**

★ T1: SYSTEMS — T2: RATIONALITY

- T1 formula: $N \subset W \subset Z \subset Q$; closed + ordered + number-line. Ex: -5 integer, 0 whole, $7/3$ rational.
- T2 formula: p/q ($q \neq 0$) + terminating/repeating; else irrational. Ex: 0.333.. rational; π irrational.

⚠ FATAL TRAP

$\sqrt{4} = 2$ (rational!) + ' $\pi = 22/7$ exactly' + 0-natural + fraction-always-rational confusions! $22/7$ approximates; $\sqrt{9}$ exits surds.

★ NORCET HIT

HIT: Prime + 1 + composites + co-prime (gcd 1!) + even/odd + parity + audit — next page drills!

► **Classify: -3, 0, $5/2$, $\sqrt{2}$, 0.4545..?**

-3 Z; 0 W; $5/2$ Q; $\sqrt{2}$ irrational; 0.4545 Q.

Prime (2 only-even!) + composite + co-prime (gcd 1!) + sieve + test-divide! Even $\pm$ even + odd $\pm$ odd + even $\times$ anything = parity predicts last-digit + sums! Ex: 97 prime; $8 \times 9 \times 5$ odd $\times$ odd $\times$ odd = odd.

2 = only-even
prime + sieve

Co-prime gcd 1
+ test-divide

Parity predicts
sums/products!

▲ T1: PRIMES — T2: PARITY

- T1 formula: prime 2 factors; test to $\sqrt{n}$; 1 neither. Ex: $91 = 7 \times 13$ composite; (8,15) co-prime.
- T2 rules: $E \pm E = E$, $O \pm O = E$, $E \pm O = O$; $E \times \text{any} = E$. Ex: odd+odd+even = even.

⚠ FATAL TRAP

1-prime + 2-composite + '91 prime' + even-prime hunt misses! Test $\sqrt{n}$ + sieve + audit.

★ NORCET HIT

HIT: Place/face + factors/multiples + remainder next — digits + division + audit!

► Prime? 1, 2, 51, 97. Odd/even? 13+17+20.

2, 97 prime; 1 neither; $51 = 3 \times 17$; sum = even.

PLACE VALUE + FACTORS/MULTIPLES

Place (position×!) + face (digit!) + expanded + compare + round! Factors (divide!) + multiples (times!) + GCF/LCM-seeds +876 = $8 \times 100 + 7 \times 10 + 6$ — digits + division + audit!

Place = digit×
position + audit

Face = digit
itself + audit

Factors÷/divide +
multiples× + audit!

◆ T1: PLACE — T2: FACTOR/MULTIPLE

- T1 formula: value = digit × place; expanded + compare + round. Ex: 7 in 876 = 70; face = 7.
- T2 formula: a|b + times-table + first-common. Ex: factors(12) = 1,2,3,4,6,12; multiples(6) = 6,12,18..

⚠ FATAL TRAP

Place↔face swap + '0 adds value' + factor/multiple-flip
+ 1-factor-everything misses! Position + divide + audit.

★ NORCET HIT

HIT: Remainder + divisibility + HCF/LCM next —
division + rules + audit!

► 7 in 5708 (place+face)? Factors of 18?

Place 700, face 7; 1,2,3,6,9,18.

Dividend = divisor × quotient + remainder ($0 \leq r$ less-than-d!) + check + verify + audit! Div-2: last-digit even (0,2,4,6,8!) + instant + audit — divide + check + last-digit!

D = d × q + r + 0 ≤ r less-than-d + audit	Div-2: last even + instant + audit	Check + verify + audit!
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Σ T1: REMAINDER — T2: DIV-BY-2

- T1 formula: $D = dq + r$; verify + remainder-theorem + audit. Ex: $47 \div 6$: q7 r5; check $6 \times 7 + 5 = 47$.
- T2 rule: units 0/2/4/6/8 → divisible. Ex: 3,458 even-yes; 7,731 no.

⚠ FATAL TRAP

$r \geq d$ -accept + 'remainder 0 = prime' + odd/even-last + no-verify = wrong + audit-fail! Divide + check + audit.

★ NORCET HIT

HIT: Div-3/4/5/6 next — sums + halves + audit!

► **Divide 100 by 7 (q+r)? Divisible by 2: 4,506?**

q14 r2 (98+2); 4,506 yes (ends 6).

DIVISIBILITY BY 3 + 4

Div-3: digit-sum $\div 3$! Div-4: last-two $\div 4$! Add + halve-twice + verify + audit — sums + pairs decide!

**Div-3: digit-sum
 $\div 3$ + audit**

**Div-4: last-two
 $\div 4$ + audit**

**Add/halve + verify
+ audit!**

★ T1: DIV-3 — T2: DIV-4

- T1 rule: sum digits; if $\div 3$ yes. Ex: 4,671 $\rightarrow$ 18 $\rightarrow$ yes; 7,732 $\rightarrow$ 19 $\rightarrow$ no.
- T2 rule: last-two $\div 4$. Ex: 3,124 (24 yes); 5,678 (78 no).

⚠ FATAL TRAP

Whole-number $\div 3$ + 'last-digit $\div 3$ ' + div-3/4-swap + no-verify = wrong + audit-fail! Sum + pair + audit.

★ NORCET HIT

HIT: Div-5/6 + combined-rules next — ends + LCM + audit!

► Div-3: 9,453? Div-4: 8,936?

9+4+5+3 = 21 yes; 36 $\div 4$ = yes.

Div-5: ends 0/5! Div-6: div-2 AND div-3 (LCM!)! Ends + sums + both + verify + audit — easy + combined rules!

**Div-5: ends 0/5
+ instant + audit**

**Div-6: div-2 AND
div-3 + audit**

**Both + verify
+ audit!**

▲ T1: DIV-5 — T2: DIV-6

- T1 rule: units 0/5. Ex: 7,890 yes; 4,567 no.
- T2 rule: even + digit-sum÷3. Ex: 4,572 (even + 18 yes); 3,471 (odd no).

⚠ FATAL TRAP

Div-6 = ÷6-last + 'either 2-or-3' + no-both + no-verify = wrong + audit-fail! Ends + sums + BOTH + audit.

★ NORCET HIT

HIT: Div-8/9 + powers-of-2/3 next — halves + nines + audit!

► **Div-5: 12,345? Div-6: 7,794?**

Ends 5 yes; even + 27 yes → yes.

Div-8: last-three ÷8 (halve-thrice!!) Div-9: digit-sum ÷9 (nines-cast-out!!) Triple-halve + nines + verify + audit — threes + nines decide!

**Div-8: last-three
÷8 + halve×3**

**Div-9: digit-sum
÷9 + cast-out**

**Verify + audit
+ pride!**

◆ T1: DIV-8 — T2: DIV-9

- T1 rule: last-three ÷8 (or halve×3). Ex: 7,336 (336÷8 = 42 yes); 5,678 no.
- T2 rule: digit-sum ÷9. Ex: 8,019 → 18 yes; 4,671 → 18 — also yes!

⚠ FATAL TRAP

Last-two÷8 + 'div-9 = div-3' + no-verify + audit-miss = wrong + fail! Three + nines + audit.

★ NORCET HIT

HIT: Div-10/11 + alternating + audit next — ends + differences!

► Div-8: 9,168? Div-9: 33,660?

168÷8 = 21 yes; sum 18 yes.

Div-10: ends 0! Div-11: (odd-sum – even-sum) ÷11 (alternating!)! Ends + alternate + difference + verify + audit — zero + zigzag decide!

**Div-10: ends 0
+ instant + audit**

**Div-11: odd–even
÷11 + audit**

**Ends/zigzag +
verify!**

Σ T1: DIV-10 — T2: DIV-11

● T1 rule: units 0. Ex: 56,780 yes; 9,876 no.

● T2 rule: odd-place sum – even-place sum ÷11. Ex: 918,082: (9+8+8)–(1+0+2) = 25–3 = 22 yes.

⚠ FATAL TRAP

Div-11 = ÷11-number + 'digits÷11' + place-mix + no-verify = wrong + audit-fail! Alternate + difference + audit.

★ NORCET HIT

HIT: Missing-digit + least-add + HCF next — algebra + division + audit!

► **Div-10: 40,905? Div-11: 2728?**

Ends 5 no; (2+2)–(7+8) = 4–15 = –11 → yes.

MISSING-DIGIT + LEAST ADD/SUBTRACT

Replace + test-0-9 + rule-fit + verify! Least = (next-multiple – number) + add/subtract + check + audit — algebra + multiples + audit!

Replace + 0-9 +
rule-fit + verify

Next-multiple –
number + audit

Algebra + multiples
+ audit!

★ T1: MISSING – T2: LEAST

- T1 method: * + rule + test digits. Ex: $4*2 \div 3 \rightarrow 4+2 = 6$, need 3/12 $\rightarrow *=1,4,7$.
- T2 formula: least = next-multiple – N. Ex: least + 476 to $\div 11$: $44 \times 11 = 484 \rightarrow +8$.

⚠ FATAL TRAP

First-digit-guess + 'any fits' + no-rule/verify + no-audit = wrong + fail! Rule + test + audit.

★ NORCET HIT

HIT: Prime-factor + division-HCF + product next — trees + ladders + audit!

► $2*5 \div 11$ (find *)? Least + 1,000 to $\div 9$?

*: $(2+5)=7$ need 4/15 $\rightarrow *=4$; $1008-1000 = +8$.

PRIME FACTORIZATION + DIVISION-HCF

Tree + divide-by-primes + exponents + ladder + verify + audit! Division-ladder + last-zero + common + HCF + audit — trees + ladders + audit!

Tree + exponents
+ ladder + audit

Division + last-zero
+ common + audit

Verify + audit
+ pride!

▲ T1: FACTOR — T2: DIVISION

- T1 method: tree/ladder + exponents. Ex: $72 = 2^3 \times 3^2$; verify multiply-back.
- T2 method: divide + remainders + HCF-last-divisor. Ex: HCF(48,180): $180 \div 48 \dots 3r36$; $48 \div 36 \dots 1r12$; $36 \div 12 = 3 \rightarrow 12$.

⚠ FATAL TRAP

1-prime + missed-exponents + 'HCF = small' + no-verify + no-audit = wrong + fail! Tree + ladder + audit.

★ NORCET HIT

HIT: Product + HCF/LCM-words next — multiply + stories + audit!

► Factor 96? HCF(36, 60) by division?

$96 = 2^5 \times 3$; $60 \div 36 \text{ r}24$; $36 \div 24 \text{ r}12$; $24 \div 12 = 2 \rightarrow 12$.

$a \times b = \text{HCF} \times \text{LCM}$ + common-divide + ratio-split + verify + audit! People/things + greatest/least + together + bells + tiles — words + HCF/LCM + audit!

$a \times b = H \times L$
+ verify + audit

Greatest/least/
together + stories

Words + audit
+ pride!

◆ T1: PRODUCT — T2: WORDS

● T1 formula: $a \times b = H \times L$ (two numbers!). Ex: H6 L72, a18 $\rightarrow b = 6 \times 72 / 18 = 24$.

● T2 map: greatest/divide = HCF; least/together = LCM. Ex: bells 12/18/20 $\rightarrow$ LCM 180 min = 3 h.

⚠ FATAL TRAP

Product-for-3-numbers + HCF/LCM-swap + 'bigger = LCM' + no-verify = wrong + fail! Map + verify + audit.

★ NORCET HIT

HIT: Fractions + decimals + BODMAS next — parts + places + order!

► H8 L60, a20 $\rightarrow$ b? Tiles 15/25 together?

$b = 8 \times 60 / 20 = 24$; $\text{LCM}(15, 25) = 75$.

FRACTION TYPES (ENTRY)

Proper + improper + mixed + like + unlike + equivalent + lowest + convert + compare + audit!
Numerator/denominator + whole + common + cross + audit — parts + wholes + audit!

Proper/improper/ mixed/like/unlike + audit	Equivalent/lowest/ convert/compare + audit	Cross + audit + pride!
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Σ TYPES — CONVERT

Type	Convert
Proper/improper/mixed	Whole + fraction + to-and-fro + audit
Like/unlike/equivalent/lowest	Common + divide + reduce + audit
All	Compare + order + operations-next + audit
⚠ FATAL TRAP Numerator/denominator-flip + 'bigger-denominator = bigger' + no-reduce + no-audit = wrong + fail! Cross + reduce + audit.	★ NORCET HIT HIT: Operations + decimals + percent next — add/sub/mul/div + places + audit!

► 7/3 to mixed? Lowest of 12/18?

2■; divide 6 → 2/3.

Add/sub (common!) + mul (straight!) + div (flip!) + decimals (align!) — common + flip + align + audit! Ex: $2/3 + 1/6 = 5/6$; $0.7 \times 0.3 = 0.21$ (2 decimals!).

Common/flip/align + audit	Straight/cross + places + audit	Ex + audit + pride!
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★ T1: FRACTIONS — T2: DECIMALS

- T1: $a/b + c/d = (ad+bc)/bd$; mul/div straight/flip. Ex: $3/4 \div 2/3 = 1/2$; $5/6 \div 6/5 = 1$.
- T2: align + count places. Ex: $2.5 + 0.75 = 3.25$; $1.2 \times 0.4 = 0.48$.

⚠ FATAL TRAP

Add-tops + 'bigger-denominator = bigger' + decimal-misalign + no-reduce = wrong + fail! Common + align + reduce.

★ NORCET HIT

HIT: Of = multiply; reciprocal + mixed + audit — next: conversions + compare!

► $3/5 + 1/10$? 0.6×0.05 ?

$7/10$; 0.030 (3 places).

CONVERSIONS + COMPARE FRACTIONS

Percent/100 + decimal-shift + fraction-reduce + cross-multiply! $1/2 = 50\%$, $1/4 = 25\%$, $1/8 = 12.5\%$
+ benchmark + audit — convert + cross + compare!

Percent/100 +
shift/reduce + audit

Cross-multiply +
benchmark + audit

Convert/compare
+ audit!

▲ T1: CONVERT — T2: COMPARE

- T1: %/100 + shift + reduce. Ex: $37.5\% = 3/8$; $0.125 = 1/8$; $2/3 = 66.67\%$.
- T2: ad vs bc; benchmarks $1/2$, $1/4$, $3/4$. Ex: $5/8$ vs $7/11$: 55 vs 56 → $7/11$ bigger.

⚠ FATAL TRAP

$37.5\% = 3.75$ + 'bigger-denominator' + no-cross +
no-reduce = wrong + fail! Shift + cross + reduce.

★ NORCET HIT

HIT: Terminating ($2/5$!) vs recurring ($1/3$, $1/6$, $1/7$!) +
BODMAS next — places + order!

► 62.5% as fraction? Bigger: $4/7$ or $5/9$?

$5/8$; 36 vs 35 → $4/7$.

Denominator 2/5-only = terminating! Brackets-Orders-Div/Mul-Add/Sub + left-to-right + audit — factors + order + audit!

**2/5-only +
 audit + pride**

**B-O-DM-AS +
 left-to-right + audit**

**Order + audit
 + pride!**

◆ T1: TERMINATE — T2: BODMAS

- T1: denominator only 2s/5s. Ex: 7/20 terminating; 5/12 recurring (3!).
- T2: B-O-DM-AS left-to-right. Ex: $12+8\div2-5 = 12+16-5 = 23$.

⚠ FATAL TRAP

Left-to-right-all + 'add first' + DM/AS-order + no-audit = wrong + fail! Brackets + orders + DM + AS.

★ NORCET HIT

HIT: Brackets + fractional-simplification + approximation next — nest + round!

► **Terminating? 9/40. Value? $20-4\times3+2$.**

Yes ($40 = 8\times5$); $20-12+2 = 10$.

BRACKETS + FRACTIONAL SIMPLIFICATION

Inner-first + vinculum + of + cancel + reduce + audit! Nested + mixed + complex + step + verify + audit — inside-out + cancel + reduce!

Inner-first +
vinculum + of + audit

Cancel + reduce
+ verify + audit

Step + audit
+ pride!

Σ T1: BRACKETS — T2: SIMPLIFY

● T1: inner + vinculum + of. Ex: $2+[3\times(4-1)] = 2+9 = 11$.

● T2: cancel + reduce + verify. Ex: $(2/3)/(4/9) = 2/3 \times 9/4 = 3/2$.

⚠ FATAL TRAP

Outer-first + of-last + no-cancel + 'simplify = decimal' + no-verify = wrong + fail! Inner + cancel + verify.

★ NORCET HIT

HIT: Approximation + exponents + roots next — round + powers!

► 5-[2+3×2]? Simplify (5/8)/(15/16)?

$5-8 = -3$; $5/8 \times 16/15 = 2/3$.

APPROXIMATION + EXPONENTS

Round + estimate + compatible + 1%/10% + audit! $a \cdot m \times a \cdot n$ + power-power + zero/one + negative + audit — round + rules + audit!

Round/estimate/ compatible + audit	$a \cdot m \times a \cdot n$ + power + $a0/a1/a\text{-neg}$ + audit	Rules + audit + pride!
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★ T1: APPROX — T2: EXPONENTS

- T1: round + compatible + 10%. Ex: $49 \times 21 \approx 50 \times 20 = 1000$ (actual 1029).
- T2: add/subtract/multiply powers; $a0 = 1$. Ex: $2^3 \times 2^2 = 2^5 = 128$; $5^0 = 1$.

⚠ FATAL TRAP

Exact-only + 'approx wrong' + power-add + $a0$ -zero + no-audit = slow + wrong + fail! Round + rules + audit.

★ NORCET HIT

HIT: Squares/cubes/roots + surds next — memorize + rationalize!

► $29 \times 12 \approx ?$ Value? $3^2 \times 3^1 = ?$

≈ 300 (348); $3^1 = 3$.

$1^2-30^2 + 1^3-12^3 + \sqrt{}/\sqrt[3]{}$ + perfect + estimate + audit! Rationalize + simplify + conjugate + audit — memorize + estimate + rationalize!

1-30 squares +
1-12 cubes + audit

Perfect/estimate +
rationalize + audit

Memory + audit
+ pride!

▲ T1: SQUARES — T2: SURDS

- T1: memorize 1-30 sq + 1-12 cubes. Ex: $17^2 = 289$; $\sqrt{144} = 12$; $\sqrt[3]{27} = 3$.
- T2: simplify + rationalize. Ex: $\sqrt{50} = 5\sqrt{2}$; $1/\sqrt{2} = \sqrt{2}/2$.

⚠ FATAL TRAP

$\sqrt{(a+b)}$ + 'root distributes' + no-memorize +
unrationalized + no-audit = wrong + slow + fail!
Memorize + rationalize + audit.

★ NORCET HIT

HIT: Ratio + proportion next — parts + wholes + audit!

► $\sqrt{169} + \sqrt[3]{64}$? Simplify $\sqrt{72}$?

$13 + 4 = 17$; $6\sqrt{2}$.

a:b + parts + total + simplify + divide + scale + audit! Equal-cross + proportion + direct/inverse + audit — parts + wholes + audit!

**a:b + parts/total
 + simplify + audit**

**Cross-equal +
 proportion + audit**

**Scale + audit
 + pride!**

◆ T1: RATIO — T2: EQUIVALENT

- T1: a:b = parts; total = sum. Ex: 24 in 2:3:4 → 16/3? parts 9: $24 \times 2/9 \approx 5.33$ —use 27: 6,9,12.
- T2: $a/b = c/d \rightarrow ad = bc$. Ex: $x/4 = 9/12 \rightarrow x = 3$.

⚠ FATAL TRAP

Add-ratio + 'ratio = fraction-value' + no-total/parts + no-verify = wrong + fail! Parts + cross + audit.

★ NORCET HIT

HIT: Direct/inverse + division + after-change next — vary + split + audit!

► **45 in 2:3:5? $x/5 = 12/20$?**

9, 13.5→20? 45: 9,13.5,22.5; $x = 3$.

DIRECT + INVERSE PROPORTION

Direct (up-up! $y = kx$!) + inverse (up-down! $xy = k$!) + table + graph + units + audit! More-more + more-less + constant + verify + audit!

Direct: $y = kx$
+ up-up + audit

Inverse: $xy = k$
+ up-down + audit

Constant + verify
+ audit!

Σ T1: DIRECT — T2: INVERSE

● T1: $y = kx$. Ex: 5 kg ■ 200 → 8 kg = $8 \times 200 / 5 = 320$.

● T2: $xy = k$. Ex: 6 men 12 days → 4 men = $6 \times 12 / 4 = 18$ days.

⚠ FATAL TRAP

Direct-always + 'more = direct' + inverse-flip + units-mix
+ no-verify = wrong + fail! Direction + constant + audit.

★ NORCET HIT

HIT: Division + after-add + compound next — split +
combine + audit!

► Direct: 3 pens 45 → 7? Inverse: 8 pipes 6 h → 12 pipes?

105; $8 \times 6 / 12 = 4$ h.

Total×parts + new-total + new-ratio + difference + constant + verify + audit! Before/after + add/remove + difference + solve + audit — split + shift + solve!

Total×parts + split + verify	Before/after + add/ remove + solve	Difference + audit!
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★ T1: DIVIDE — T2: CHANGE

● T1: share = total × part/sum. Ex: 550 in 3:5:3 → 150, 250, 150.

● T2: before/after + difference constant. Ex: ratio 4:5, +20 each → 6:7? solve: $(4x+20)/(5x+20) = 6/7 \rightarrow x = 20 \rightarrow 80,100$.

⚠ FATAL TRAP

Old-total + 'ratio stays' + difference-ignored + no-verify
 = wrong + fail! New-total + constant + audit.

★ NORCET HIT

HIT: Compound + percent-bank next — combine + convert!

► 600 in 1:2:3? Ratio 2:3, +10 each → 3:4, find?

100,200,300; $(2x+10)/(3x+10)=3/4 \rightarrow x=10 \rightarrow 20,30$.

Multiply + simplify + part-whole + audit! 1 = 100%, 1/2 = 50%, 1/4 = 25%, 1/5 = 20%, 1/8 = 12.5%, 1/3, 2/3, 1/6, 1/7, 1/9, 1/11, 1/12 + audit — combine + convert!

**Multiply + simplify
 + audit + pride**

**Bank + benchmarks
 + instant + audit**

**Tables + audit
 + pride!**

▲ T1: COMPOUND — T2: BANK

● T1: (a:b)(c:d) = ac:bd. Ex: (2:3)(4:5) = 8:15.

● T2 bank: 1/2 50, 1/4 25, 1/8 12.5, 1/3 33.33, 2/3 66.67, 1/6 16.67, 1/7 14.28, 1/9 11.11, 1/11 9.09.

⚠ FATAL TRAP

Add-ratios + 'compound = average' +
 bank-unmemorized + no-audit = slow + wrong + fail!
 Multiply + memorize + audit.

★ NORCET HIT

HIT: Basic + increase/decrease next — of + change +
 audit!

► **(3:4)(5:7)? 37.5% + 12.5% as fractions?**

15:28; 3/8 + 1/8 = 1/2.

BASIC + INCREASE/DECREASE PERCENT

of = multiply + change/original×100 + new/old + audit! Increase + decrease + profit + loss + growth + shrink + verify + audit!

of = multiply
+ audit + pride

Change/original×
100 + audit

New/old + verify
+ audit!

◆ T1: BASIC — T2: CHANGE

- T1: $p\%$ of $N = p \times N / 100$. Ex: 15% of 240 = 36; 33.33% of 90 = 30.
- T2: $\text{change}\% = \text{diff} / \text{original} \times 100$. Ex: $80 \rightarrow 100 = +25\%$; $120 \rightarrow 96 = -20\%$.

⚠ FATAL TRAP

Percent-of-wrong-base + 'increase = new' + base-mix + no-verify = wrong + fail! Base + verify + audit.

★ NORCET HIT

HIT: Successive + reverse + points next — chain + back + distinguish!

► 20% of 350? $50 \rightarrow 65$ change%?

70; +30%.

SUCCESSIVE + REVERSE PERCENT

a+b+ab/100 + net + chain + audit! Original = final×100/(100±r) + back + verify + audit — chain + backtrack!

**a+b+ab/100 +
net + chain + audit**

**Back: final×100/
(100±r) + audit**

**Verify + audit
+ pride!**

Σ T1: SUCCESSIVE — T2: REVERSE

● T1: net = a+b+ab/100 (signs!). Ex: +20 −10: 20−10−2 = +8%.

● T2: O = F×100/(100±r). Ex: after +25% = 500 → O = 500×100/125 = 400.

⚠ FATAL TRAP

Add-percent + 'net = sum' + reverse-forward + ab-sign
+ no-verify = wrong + fail! Signs + back + audit.

★ NORCET HIT

HIT: Points + population + average next — distinguish
+ people + mean!

► **Net: +10 +10? After −20% = 320, original?**

+21%; 320×100/80 = 400.

POINTS VS CHANGE + POPULATION %

Points (subtract!) vs percent (divide by base!) + increase + decrease + people + income + spend + growth + audit! Points-difference + base-divide + verify + audit!

Points = subtract
+ audit + pride

Percent = base-
divide + audit

People/income +
verify!

★ T1: POINTS — T2: POPULATION

- T1: points = new% – old%. Ex: 40% → 46% = +6 points (NOT +15%).
- T2: people/income/spend + growth + verify. Ex: pop 50k +12%: $50k \times 1.12 = 56k$.

⚠ FATAL TRAP

Points-as-percent + '+6 points = +6%' + base-mix + no-verify = wrong + fail! Subtract vs divide + audit.

★ NORCET HIT

HIT: Averages next — sum/count + missing + combined + weighted + audit!

► 45% → 54%? Pop 80k –15%?

+9 points (+20%); 68k.

Sum/count + total + back-calculate + verify + audit! $\text{Avg} \times n = \text{total} + \text{miss} + \text{check} + \text{audit} - \text{total} + \text{back} + \text{verify}!$

Sum/count + total + audit	Back: $\text{avg} \times n +$ known + verify	Check + audit + pride!
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▲ T1: SIMPLE — T2: MISSING

- T1: $\text{avg} = \text{sum}/n$. Ex: 12,18,20,25,25 $\rightarrow 100/5 = 20$.
- T2: $\text{miss} = \text{avg} \times n - \text{known}$. Ex: avg 5 nums 20, known sum 82 $\rightarrow$ miss 18.

⚠ FATAL TRAP

Divide-count-wrong + 'average = middle' + miss-guess
 + no-verify = wrong + fail! Sum + back + audit.

★ NORCET HIT

HIT: Combined + weighted + speed next — mix +
 weights + harmonic!

► Avg 8 nums = 15, 7 sum 96. Miss?

Total 120 – 96 = 24.

COMBINED + WEIGHTED AVERAGE

Total/total-n + weight×value/sum-weights + mix + verify + audit! Groups + credits + prices + ratios + audit — totals + weights + audit!

Totals/total-n + verify + audit	Weight×value/sum + mix + audit	Verify + audit + pride!
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◆ T1: COMBINED — T2: WEIGHTED

- T1: $(n1a1+n2a2)/(n1+n2)$. Ex: $30@40 + 20@55 = (1200+1100)/50 = 46$.
- T2: $\text{sum}(w \times x) / \text{sum}(w)$. Ex: $70 \times 3 + 90 \times 1 / 4 = 300/4 = 75$.

⚠ FATAL TRAP

Simple-mean-of-means + 'average averages' + weights-ignored + no-verify = wrong + fail! Totals + weights + audit.

★ NORCET HIT

HIT: Speed + CP/SP/MP next — harmonic + prices + audit!

► **40@60 + 60@90 combined? 2:3 weights 50/80?**

$7800/100 = 78$; $(100+240)/5 = 68$.

Total-dist/total-time (harmonic!) + cost + sell + mark + list + audit! $2xy/(x+y)$ + CP/SP/MP + profit + audit — harmonic + prices + audit!

$2xy/(x+y)$ + total/total + audit	CP/SP/MP + list + audit	Verify + audit + pride!
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Σ T1: SPEED — T2: PRICES

- T1: $2xy/(x+y)$ equal-dist. Ex: $40 + 60 \rightarrow 2 \times 40 \times 60 / 100 = 48$ (not 50!).
- T2: CP + MP + SP + list. Ex: CP500 MP800 SP680: disc = $120/800 = 15\%$.

⚠ FATAL TRAP

Arithmetic-mean-speed + 'CP = SP' + MP/SP-swap + no-verify = wrong + fail! Harmonic + prices + audit.

★ NORCET HIT

HIT: P&L; + discount + successive next — percent + chain + audit!

► **60 + 90 equal-dist? CP300 SP375 profit%?**

$2 \times 60 \times 90 / 150 = 72$; $75/300 = 25\%$.

PROFIT/LOSS + DISCOUNT

P&L; on CP! + SP/CP×100 + discount on MP! + list + audit! Gain + loss + deal + verify + audit — base + percent + audit!

P&L; on CP + SP/CP + audit	Discount on MP + list + audit	Verify + audit + pride!
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★ T1: P&L; — T2: DISCOUNT

- T1: $P\% = (SP - CP) / CP \times 100$. Ex: CP200 SP250 = +25%; SP170 = -15%.
- T2: $D\% = (MP - SP) / MP \times 100$. Ex: MP1000 SP850 = 15%.

⚠ FATAL TRAP

P&L-on-SP; + discount-on-CP + 'profit = SP' +
base-mix + no-verify = wrong + fail! CP vs MP + audit.

★ NORCET HIT

HIT: Successive + MP-profit + false-weight next —
chain + deal + cheat!

► CP400 SP460? MP2000 SP1700 disc%?

+15%; $300 / 2000 = 15\%$.

a+b+ab/100 + net + chain + MP + CP + SP + margin + verify + audit! Deal + chain + profit + audit — chain + margin + audit!

**a+b+ab/100 +
 net + chain + audit**

**MP/CP/SP + margin
 + verify + audit**

**Verify + audit
 + pride!**

▲ T1: SUCCESSIVE — T2: MP-PROFIT

● T1: net = a+b+ab/100. Ex: 20+10: 20+10+2 = 32%? No: -20 -10: -20-10+2 = -28%.

● T2: want 20% on CP600, MP800: SP = 720; disc = 80/800 = 10%.

⚠ FATAL TRAP

Add-discounts + 'net = sum' + MP/CP-mix +
 margin-miss + no-verify = wrong + fail! Signs + margin
 + audit.

★ NORCET HIT

HIT: False-weight + SI + P/R/T next — cheat + interest
 + audit!

► **Net: -15 -10? CP500 MP650, want +30%?**

-15-10+1.5 = -23.5%; SP650? 500×1.3=650 → 0% disc.

FALSE WEIGHT + SIMPLE INTEREST

Claim/actual + gain% + cheat + verify + audit! PRT/100 + amount + rate + time + verify + audit —
cheat + interest + audit!

Claim/actual + gain% + audit	PRT/100 + amount + verify + audit	Verify + audit + pride!
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◆ T1: FALSE — T2: SI

- T1: gain% = (claim–actual)/actual×100. Ex: 1000 vs 950: 50/950 = 5.26%.
- T2: SI = PRT/100; A = P+SI. Ex: 5000, 8%, 3y = 1200; A = 6200.

⚠ FATAL TRAP

Gain-on-claim + 'SI = compound' + rate/time-mix +
no-verify = wrong + fail! Actual + simple + audit.

★ NORCET HIT

HIT: Find-P/R/T + SI-words + compounding next —
back + stories + growth!

► 950 for 1000 gain%? SI: 8000, 5%, 4y?

50/950 = 5.26%; 1600.

FIND P/R/T + SI WORD PROBLEMS

P = SI×100/RT + R + T + back + verify + audit! Installment + shortfall + double + compare + stories + audit — backtrack + stories + audit!

**Back: SI×100/RT
+ verify + audit**

**Installment/shortfall/
double/compare + audit**

**Stories + audit
+ pride!**

Σ T1: BACK — T2: WORDS

● T1: P/R/T from SI×100/RT. Ex: SI900, 6%, 3y → P = 900×100/18 = 5000.

● T2: double/mix/compare + stories. Ex: double in 8y → R = 100/8 = 12.5%.

⚠ FATAL TRAP

SI-for-P + rate/time-flip + 'double = 2×rate' + no-back + no-verify = wrong + fail! Back + stories + audit.

★ NORCET HIT

HIT: Compounding (annual/half) + difference next — growth + compare!

► **SI600, 5%, 4y → P? Triple in 20y → R?**

P = 3000; R = 200/20 = 10%.

ANNUAL + HALF-YEARLY COMPOUNDING

A = $P(1+r/100)^n$ + rate-half + time-double + effective + verify + audit! CI = A–P + growth + compare + audit — power + periods + audit!

**A = $P(1+r/100)^n$
+ verify + audit**

**Half: $r/2 + 2n$
+ effective + audit**

**Power + audit
+ pride!**

★ T1: ANNUAL — T2: HALF

● T1: $A = P(1+r/100)^n$. Ex: 10000, 10%, 2y = $10000 \times 1.21 = 12100$; CI = 2100.

● T2: $r/2, 2n$. Ex: 10000, 10%, 1y half-yearly: $10000 \times (1.05)^2 = 11025$.

⚠ FATAL TRAP

SI-formula + 'half = half-SI' + rate/time-unadjusted + no-power + no-verify = wrong + fail! Adjust + power + audit.

★ NORCET HIT

HIT: SI-vs-CI + growth + partnership next — compare + share + audit!

► CI: 5000, 8%, 2y? Half-yearly 1y?

$5000 \times 1.1664 = 5832 \rightarrow 832$; $5000 \times (1.04)^2 = 5408$.

SI-CI DIFFERENCE + GROWTH/DEPRECIATION

2-yr: $PR^2/100^2$! + 3-yr + growth + decay + population + machines + audit! Difference + appreciate + depreciate + verify + audit!

2-yr $PR^2/100^2$
+ verify + audit

Growth/decay +
population + audit

Verify + audit
+ pride!

▲ T1: DIFF — T2: GROWTH

- T1: 2-yr diff = $PR^2/100^2$. Ex: 10000, 10%: diff = 100.
- T2: $P(1 \pm r/100)^n$. Ex: pop 80000 +5% 2y = $80000 \times 1.1025 = 88200$.

⚠ FATAL TRAP

CI-SI-guess + 'growth = SI' + sign-mix + no-verify =
wrong + fail! Formula + signs + audit.

★ NORCET HIT

HIT: Partnership (capital-time) next — share + join +
salary!

► Diff: 4000, 5%, 2y? Decay 10% on 50000, 2y?

$4000 \times 25/10000 = 10$; $50000 \times 0.81 = 40500$.

CAPITAL-TIME + JOIN/LEAVE PARTNERS

Share = $C \times T$ + ratio + total + verify + audit! Join/leave + months + salary/commission + pre-share + audit — money $\times$ months + verify!

Share = $C \times T$
+ ratio + audit

Join/leave + months
+ salary + audit

Verify + audit
+ pride!

◆ T1: RATIO — T2: JOIN

- T1: ratio C1T1:C2T2. Ex: $5000 \times 12 : 8000 \times 6 = 60000:48000 = 5:4$; profit 9000 $\rightarrow$ 5000,4000.
- T2: join/leave + salary-first. Ex: A 12mo, B 8mo + B salary 500, profit 6500: ratio first, then split rest.

⚠ FATAL TRAP

Capital-only + 'equal split' + time-ignored + salary-last + no-verify = wrong + fail! Money $\times$ months + salary-first + audit.

★ NORCET HIT

HIT: Weighted-average + alligation next — mix + rule + audit!

► **6000 $\times$ 9 : 9000 $\times$ 6, profit 13000?**

54000:54000 = 1:1 $\rightarrow$ 6500 each.

SALARY + WEIGHTED AVERAGE (PARTNERS)

Salary/commission-first + rest-ratio + verify + audit! Weight×value/sum + mix + verify + audit — salary-first + weights + audit!

Salary-first +
rest-ratio + audit

Weight×value/sum +
mix + verify

Audit + pride
+ date!

Σ T1: SALARY — T2: WEIGHTED

- T1: salary-first, split rest. Ex: profit 10000, B salary 1000: rest 9000 in 2:1 → A6000, B1000+3000.
- T2: $\text{sum}(w \times x) / \text{sum}(w)$. Ex: marks 3:2, 60/90: $(180+180)/5 = 72$.

⚠ FATAL TRAP

Split-then-salary + 'average averages' + weights-ignored + no-verify = wrong + fail! Salary-first + weights + audit.

★ NORCET HIT

HIT: Alligation + replacement + dilution next — rule + swap + water!

► Profit 13000, A salary 1000, ratio 1:1 rest? Weights 1:4, 40/100?

Rest 12000 → 6000 + (1000+6000); $(40+400)/5 = 88$.

WEIGHTED AVG + ALLIGATION

sum(w×x)/sum(w) + cheaper/dearer/mean + difference-ratio + verify + audit! Mix + price + ratio + quantity + audit — weights + rule + audit!

**sum(w×x)/sum(w)
+ verify + audit**

**Cheap/dear/mean +
diff-ratio + audit**

**Mix + rule
+ audit!**

★ T1: WEIGHTED — T2: ALLIGATION

● T1: $\text{sum}(w \times x) / \text{sum}(w)$. Ex: $60 \times 2 + 90 \times 3 / 5 = 390 / 5 = 78$.

● T2: $(\text{dear} - \text{mean}) : (\text{mean} - \text{cheap})$. Ex: 12 + 20 for 15: $(20 - 15) : (15 - 12) = 5 : 3$.

⚠ FATAL TRAP

Simple-mean + 'alligation always' + ratio-flip + no-verify
= wrong + fail! Weights + order + audit.

★ NORCET HIT

HIT: Replacement + dilution next — swap + water +
audit!

► **Rs.30×2kg + Rs.50×3kg avg? Mix 10 + 25 for 20?**

$210 / 5 = 42$; $(25 - 20) : (20 - 10) = 5 : 10 = 1 : 2$.

REPLACEMENT + DILUTION

Final = initial × ((total – out) / total)ⁿ + water + wine + milk + audit! Swap + repeat + power + verify + audit — replace + power + audit!

**Final = init × ((T – o) / T)ⁿ
 + verify**

**Water/wine/milk +
 swap + audit**

**Power + audit
 + pride!**

▲ T1: REPLACE — T2: DILUTE

- T1: $F = I \times ((T - o) / T)^n$. Ex: 100L, out 10 twice: $100 \times (0.9)^2 = 81$ L milk.
- T2: add-water + ratio + price + verify. Ex: 1:3 milk:water, price 1/4th.

⚠ FATAL TRAP

Single-step + 'linear' + power-miss + no-verify = wrong
 + fail! Power + repeat + audit.

★ NORCET HIT

HIT: Ages + work next — translate + rate + audit!

► 80L, out 20 twice. Left?

$$80 \times (0.75)^2 = 45\text{L.}$$

Present + past/future + difference-constant + sum + ratio + verify + audit! x + years + shift + solve + audit — translate + constant + audit!

**Present/past/future
+ shift + solve**

**Difference-constant
+ sum + ratio**

**Verify + audit
+ pride!**

◆ T1: TRANSLATE — T2: RATIOS

● T1: now x, +n/−n + shift. Ex: A now 20, B 12; 5y ago ratio? 15:7.

● T2: difference constant + sum. Ex: ratio 3:5, sum 40 → 15, 25.

⚠ FATAL TRAP

Present-only + 'ratio stays' + difference-shift + no-verify
= wrong + fail! Shift + constant + audit.

★ NORCET HIT

HIT: Family-average + work-rate next — mean + rate + audit!

► Father/son 5:1, diff 24. Ages?

$5x - x = 24 \rightarrow x6: 30, 6.$

Total/n + new-born/marriage + shift + verify + audit! 1/a + 1/b + combined + rate + audit — mean + rate + audit!

**Total/n + shift
 + verify + audit**

**1/a+1/b + combined
 + rate + audit**

**Verify + audit
 + pride!**

Σ T1: FAMILY — T2: RATE

- T1: total±shift + verify. Ex: avg 5 of 30 → total 150.
- T2: combined = $ab/(a+b)$ for 2. Ex: 6 + 12 days → $72/18 = 4$ days.

⚠ FATAL TRAP

Average-only + 'add rates' + shift-miss + no-verify = wrong + fail! Total + reciprocal + audit.

★ NORCET HIT

HIT: Combined + efficiency + men-days next — together + ratio + audit!

► **A 10d + B 15d together? Family avg 4 of 20?**

150/25 = 6d; total 80.

Together + ratio + parts + wages + verify + audit! $A = 2B$ + half + double + days + audit —
 together + ratio + audit!

Together + ratio +
 parts + wages

$2B$ /half/double +
 days + audit

Verify + audit
 + pride!

★ T1: COMBINED — T2: EFFICIENCY

● T1: $1/a+1/b+1/c$; wages ratio. Ex: $A+B$ 12d, wages 9000 in 1:2 → 3000, 6000.

● T2: $A = 2B \rightarrow B = 2A$ days. Ex: A twice B, together 12d: A 18d, B 36d.

⚠ FATAL TRAP

Add-days + 'equal wages' + efficiency-flip + no-verify =
 wrong + fail! Reciprocal + ratio + audit.

★ NORCET HIT

HIT: Men-days + alternate + wages next — hours +
 days + pay!

► $A+B$ 10d, $A-B$ 1:2 wages 6000?

2000, 4000.

MEN-DAYS-HOURS + ALTERNATE DAYS

M1D1H1/W1 = M2D2H2/W2 + pairs + verify + audit! A+B + B+C + 2-day-cycles + left + verify + audit — pairs + cycles + audit!

**M1D1H1/W1 =
M2D2H2/W2 + audit**

**2-day + left +
cycles + verify**

**Pairs + audit
+ pride!**

▲ T1: MDH — T2: ALTERNATE

- T1: pairs equal. Ex: 12 men 8d = 16 men x $\rightarrow$ x = 6d.
- T2: 2-day work + remainder. Ex: A+B 2d = $1/6 + 1/12$? A6d B12d alternate A-first: 2d = $1/4$; 4 cycles = 8d + ...

⚠ FATAL TRAP

Men-only + 'alternate = half' + remainder-miss + who-first + no-verify = wrong + fail! Pairs + cycles + audit.

★ NORCET HIT

HIT: Wages + pipes next — pay + flow + audit!

► 20 men 12d = 30 men? A5d B10d alternate B-first, total?

8d; Per 2d (B+A) = $3/10$; 3 cycles = $9/10$ in 6d; day-7 B finishes $\rightarrow$ 7 days.

Wages-work-ratio + inverse-days + verify + audit! Fill/empty + net + time + verify + audit — pay + flow + audit!

**Wages-work-ratio
 + verify + audit**

**Fill/empty/net +
 time + audit**

**Verify + audit
 + pride!**

◆ T1: WAGES — T2: INLET

● T1: pay = work ratio. Ex: A 8d B 12d, 5000 → 3000, 2000.

● T2: net rate + time. Ex: fill 6h + empty 10h: net $1/6 - 1/10 = 1/15$ → 15h.

⚠ FATAL TRAP

Equal-pay + 'inlet only' + net-miss + no-verify = wrong + fail! Ratio + net + audit.

★ NORCET HIT

HIT: Leak + pipes-times next — hole + schedule + audit!

► A6d B12d 9000? Fill 4h empty 12h net?

6000, 3000; $1/4 - 1/12 = 1/6$ → 6h.

LEAK + PIPES SCHEDULE

Leak = negative-rate + extra-time + verify + audit! Open/close + combined + left + verify + audit — hole + schedule + audit!

Leak-negative +
extra-time + audit

Open/close + left +
combined + verify

Audit + pride
+ date!

Σ T1: LEAK — T2: SCHEDULE

- T1: leak slows + extra. Ex: fill 8h, leak +2h → net $1/8 - 1/10 = 1/40$ → empties 40h.
- T2: schedule + left + verify. Ex: A+B 2h, A leaves, B alone 6h → ...

⚠ FATAL TRAP

Leak-ignored + 'fill rate' + schedule-miss + no-verify = wrong + fail! Negative + left + audit.

★ NORCET HIT

HIT: Speed + conversion + relative next — dist/time + units + audit!

► Fill 6h, with leak 8h. Leak alone?

$1/6 - 1/8 = 1/24 \rightarrow 24h$.

D = S×T + km/h×5/18 = m/s + back + verify + audit! Units + back-convert + verify + audit — dist/time + units + audit!

**D = S×T +
verify + audit**

**km/h×5/18=m/s
+ back + audit**

**Units + audit
+ pride!**

★ T1: SDT — T2: CONVERT

● T1: D = ST. Ex: 72 km/h 25 min = 72×25/60 = 30 km.

● T2: ×5/18, ×18/5. Ex: 90 km/h = 25 m/s; 20 m/s = 72 km/h.

⚠ FATAL TRAP

Time-units-mix + 'km/h = m/s' + factor-flip + no-verify = wrong + fail! Units + factor + audit.

★ NORCET HIT

HIT: Relative + meeting + average-speed next — approach + together + harmonic!

► **54 km/h 40 min dist? 15 m/s in km/h?**

36 km; 54 km/h.

Same/opposite + sum/difference + meet/catch + time + verify + audit! Together + apart + overtake + audit — approach + catch + audit!

Same/opposite +
 sum/difference + audit

Meet/catch + time
 + verify + audit

Audit + pride
 + date!

▲ T1: RELATIVE — T2: MEET

- T1: same –, opposite +. Ex: 60 + 40 opposite = 100 km/h approach.
- T2: meet = dist/sum; catch = dist/diff. Ex: 300 km, 50+70 → 2.5 h.

⚠ FATAL TRAP

Always-add + 'meet halfway' + direction-miss +
 no-verify = wrong + fail! Direction + formula + audit.

★ NORCET HIT

HIT: Average-speed + trains next — harmonic + length
 + audit!

► A60 B90 same-dir behind 120 km. Catch time?

Diff 30 → 4 h.

AVG SPEED (UNEQUAL) + TRAIN-POLE

Total/total-time + equal-harmonic + train+object/speed + verify + audit! Length + speed + time + audit — harmonic + length + audit!

Total/total-time +
harmonic + audit

(Train+obj)/speed
+ verify + audit

Audit + pride
+ date!

◆ T1: AVG — T2: POLE

- T1: total/total-time. Ex: $60@2h + 90@1h = 210/3 = 70$.
- T2: length/speed. Ex: 240m train, 72 km/h = 20 m/s → 12 s.

⚠ FATAL TRAP

Simple-mean + 'train point' + length-miss + no-verify = wrong + fail! Total + length + audit.

★ NORCET HIT

HIT: Platform + two-trains next — lengths + opposite + audit!

► 30@1h + 90@2h avg? 180m, 54 km/h pole?

$210/3 = 70$; 15 m/s → 12 s.

Train+platform/speed + opposite/same + sum/difference + meet + verify + audit! Lengths + directions + audit — lengths + approach + audit!

Train+platform/
 speed + verify

Opposite/same +
 sum/difference + audit

Audit + pride
 + date!

Σ T1: PLATFORM — T2: TWO

● T1: $(T+P)/S$. Ex: $150+250\text{m}$, $72\text{ km/h} = 20\text{ m/s} \rightarrow 20\text{ s}$.

● T2: $(L1+L2)/(v1 \pm v2)$. Ex: $120+180\text{m}$, $60+90\text{ opposite km/h} = 41.67\text{ m/s} \rightarrow 7.2\text{ s}$.

⚠ FATAL TRAP

Train-only + 'same = add' + units-mix + no-verify =
 wrong + fail! Lengths + direction + units + audit.

★ NORCET HIT

HIT: Boats + clocks + calendar next — stream + hands
 + days!

► **200+300m, 90 km/h opposite 54? Cross time?**

Sum $144\text{ km/h} = 40\text{ m/s}$; $500/40 = 12.5\text{ s}$.

Down = still+stream! Up = still–stream! Solve-pair + times + verify + audit — add/subtract + solve!

Down = $b+w$
Up = $b-w$ + audit

Solve-pair + times
+ verify + audit

Audit + pride
+ date!

★ T1: RELATION — T2: SPEEDS

- T1: $D = b+w$, $U = b-w$. Ex: D15 U9 → b12 w3.
- T2: $b = (D+U)/2$, $w = (D-U)/2$. Ex: D20 U12 → b16 w4.

⚠ FATAL TRAP

Add-for-upstream + 'still = average always' + no-pair/solve + no-verify = wrong + fail! Add/subtract + solve + audit.

★ NORCET HIT

HIT: Boat-times + clocks + calendar next — schedule + hands + days!

► D18 U10. b? w?

b14, w4.

Time = dist/speed + upstream/downstream + verify + audit! 30°/hour + 0.5°/min + 6°/min + relative + verify + audit — schedule + angles + audit!

**Time = dist/speed
 + verify + audit**

**30°/h + 0.5°/m
 + 6°/m + audit**

**Verify + audit
 + pride!**

▲ T1: TIMES — T2: HANDS

● T1: $t = D/S$ each-way. Ex: 60 km, D15 U10: $4+6 = 10$ h.

● T2: hour 30°/h, min 6°/m. Ex: 3:00 → 90°.

⚠ FATAL TRAP

Speed-add + 'clock = 360/12 only' + minute-miss + no-verify = wrong + fail! Each-way + both-hands + audit.

★ NORCET HIT

HIT: Angle + coincide + gain/loss next — degrees + meet + correct!

► **72 km, D18 U12 round time? Angle 6:00?**

$4+6 = 10$ h; 180°.

Angle = $|30H - 5.5M| + \text{mod-}360 + \text{verify} + \text{audit!}$ Coincide/opposite/right-angle + 65-5/11-min + verify + audit — formula + frequency + audit!

$|30H - 5.5M| +$
mod-360 + audit

65-5/11 min +
meet + verify

Formula + audit
+ pride!

◆ T1: ANGLE — T2: MEET

- T1: $|30H - 5.5M|$, mod 360. Ex: 4:20 $\rightarrow |120 - 110| = 10^\circ$.
- T2: coincide every 65-5/11 min ($\sim 11 \times /12h$). Ex: between 7-8 $\approx 7:38$.

⚠ FATAL TRAP

Hour-only + 'coincide hourly' + mod-miss + no-verify = wrong + fail! Both-hands + frequency + audit.

★ NORCET HIT

HIT: Gain/loss + leap + odd-days next — correct + rules + count!

► Angle 8:24? Coincide count per 12h?

$|240 - 132| = 108^\circ$; 11 times.

Fast/slow + per-day + correct-time + verify + audit! Divisible-4 + century-400 + verify + audit — correct + rules + audit!

**Fast/slow +
per-day + correct**

**Div-4 + century-
400 + verify**

**Audit + pride
+ date!**

Σ T1: GAIN — T2: LEAP

● T1: gain/loss per day + correct. Ex: +5 min/day, after 12h shown? actual 11:55... (12h: gain 5/12 min... keep simple: set right daily).

● T2: div-4, century div-400. Ex: 2000 leap; 1900 not; 2024 leap.

⚠ FATAL TRAP

Gain-as-correct + 'century always' + 400-miss + no-verify = wrong + fail! Correct + rules + audit.

★ NORCET HIT

HIT: Odd-days + day-calc + repeat next — count + anchor + cycle!

► Leap? 1800, 2000, 2028?

No (century/400); yes; yes.

Year/100/400 + month + date + mod-7 + Sunday-Saturday + verify + audit! Remainders + anchors + doomsday + audit — count + mod + audit!

Year/100/400 + month/date + audit	Mod-7 + anchor + verify + audit	Audit + pride + date!
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★ T1: ODD — T2: DAY

- T1: odd = remainder mod 7. Ex: 100 yrs = 5; 400 yrs = 0; ordinary yr = 1; leap yr = 2.
- T2: total odd + weekday. Ex: 1 Jan 2000 = Saturday (known anchor) → practice with given anchor.

⚠ FATAL TRAP

All-months-30 + 'odd = even' + leap-miss +
 anchor-miss + no-verify = wrong + fail! Remainders +
 anchors + audit.

★ NORCET HIT

HIT: Calendar-repeat + linear-equations next — cycle +
 algebra + audit!

► Odd days in 350 days? Ordinary year odd?

$350 \bmod 7 = 0; 1.$

CALENDAR REPEAT + LINEAR (ONE VAR)

28/40 + leap/ordinary + Jan/Feb-leap + verify + audit! $ax+b = c$ + isolate + check + verify + audit — cycle + isolate + audit!

28/40 + leap +
Jan/Feb + verify

$ax+b = c$ +
isolate + check

Cycle + audit
+ pride!

▲ T1: REPEAT — T2: LINEAR

- T1: leap +28, ordinary +11/+6 cycle. Ex: 2024 leap → 2052.
- T2: isolate + check. Ex: $3x+7 = 25 \rightarrow x = 6$; check 25.

⚠ FATAL TRAP

Repeat-guess + 'same date' + leap-miss + sign-flip + no-check = wrong + fail! Cycle + isolate + audit.

★ NORCET HIT

HIT: Two-equations + identities next — solve + expand + audit!

► 2023 repeats? Solve $5x-11 = 24$?

2034 (+11); $x = 7$.

TWO LINEAR EQUATIONS + IDENTITIES

Eliminate/substitute + verify + audit! $(a+b)^2 + (a-b)^2 + a^2 - b^2$ + expand + verify + audit — solve + expand + audit!

Eliminate/substitute
+ verify + audit

Squares/difference +
expand + verify

Audit + pride
+ date!

◆ T1: SOLVE — T2: IDENTITIES

- T1: eliminate/substitute. Ex: $x+y = 10$, $x-y = 4 \rightarrow x6, y4$.
- T2: $(a+b)^2$, $a^2 - b^2 = (a-b)(a+b)$. Ex: $103^2 - 97^2 = 10000 - 9 = 9991$.

⚠ FATAL TRAP

Add-all + 'solution obvious' + identity-mix + no-verify = wrong + fail! Eliminate + expand + audit.

★ NORCET HIT

HIT: Word-translation + squares next — English + areas + audit!

► $2x+3y = 17$, $x-y = 1$? 98·102?

$x4$? $2x+3y=17$, $x = y+1 \rightarrow 5y = 15 \rightarrow y3, x4$; $10000-4 = 9996$.

English-to-algebra + let + relate + solve + check + audit! Side² + diagonal + perimeter + verify + audit — translate + square + audit!

Let/relate/solve/
check + audit

Side² + diagonal +
perimeter + audit

Verify + audit
+ pride!

Σ T1: WORDS — T2: SQUARE

- T1: let + relate + solve + check. Ex: 5 more than twice x is 25 → $x = 10$.
- T2: $A = s^2$, $P = 4s$, $d = s\sqrt{2}$. Ex: P36 → s9, A81.

⚠ FATAL TRAP

Numbers-first + 'of = multiply always' + units-mix + no-check = wrong + fail! Translate + check + audit.

★ NORCET HIT

HIT: Rectangle + triangle next — length×width + half-base-height!

► 'Thrice less 4 is 20'? Square P28 area?

$3x - 4 = 20 \rightarrow x8$; $s7 \rightarrow 49$.

RECTANGLE + TRIANGLE

$L \times W + 2(L+W)$ + diagonal + verify + audit! $\text{Half} \times \text{base} \times \text{height}$ + Pythagoras + Heron + verify + audit — length + half + audit!

**$L \times W + 2(L+W)$
+ diagonal + audit**

**$\text{Half} \times b \times h$ + Pyth +
Heron + audit**

**Verify + audit
+ pride!**

★ T1: RECT — T2: TRI

● T1: $A = LW$, $P = 2(L+W)$. Ex: 12×5 : A60, P34.

● T2: $A = bh/2$; 3-4-5; Heron. Ex: 6,8,10 $\rightarrow$ 24.

⚠ FATAL TRAP

$L+W$ -area + 'slant = height' + Pythagoras-mix + units + no-verify = wrong + fail! Length + half + audit.

★ NORCET HIT

HIT: Equilateral + circle next — root-3/4 + pi-r-squared!

► **15×8 area/perimeter? Triangle 5,12,13 area?**

120, 46; 30.

Root- $\frac{3}{4} \times \text{side}^2$ + height + verify + audit! πr^2 + 2- πr + diameter + verify + audit — root + π + audit!

$\sqrt{\frac{3}{4} \times a^2}$ + height
+ verify + audit

πr^2 + 2- πr
+ diameter + audit

Verify + audit
+ pride!

▲ T1: EQUILATERAL — T2: CIRCLE

● T1: $A = \sqrt{\frac{3}{4}} a^2$, $h = \sqrt{\frac{3}{2}} a$. Ex: $a6 \rightarrow 9\sqrt{3} \approx 15.6$.

● T2: $A = \pi r^2$, $C = 2\pi r$. Ex: $r7 \rightarrow A154$, $C44$.

⚠ FATAL TRAP

Base $\times$ height + 'pi = 3.14 always' + diameter-radius-mix
+ no-verify = wrong + fail! Root + pi + audit.

★ NORCET HIT

HIT: Semicircle + composite next — half + add/subtract
+ audit!

► Equilateral 4 area? Circle r14 (22/7)?

$4\sqrt{3} \approx 6.9$; A616, C88.

Half-circle + half-perimeter + diameter + verify + audit! Add + subtract + shaded + paths + verify + audit — half + combine + audit!

Half-area/perimeter
+ diameter + audit

Add/subtract/shaded/
paths + verify

Audit + pride
+ date!

◆ T1: SEMI — T2: COMPOSITE

● T1: $A = \pi r^2/2$, $P = \pi r + 2r$. Ex: $r7 \rightarrow A77, P36$.

● T2: add/subtract + shaded + paths. Ex: square 14 minus circle $r7$: $196 - 154 = 42$.

⚠ FATAL TRAP

Full-circle + 'shaded = total' + diameter-miss + no-verify
= wrong + fail! Half + subtract + audit.

★ NORCET HIT

HIT: Conversions + cube next — units + volume +
audit!

► Semicircle $r10$ (3.14)? Shaded: 10-square minus $r5$ circle?

A157, P51.4; $100 - 78.5 = 21.5$.

UNIT CONVERSIONS (LENGTH/AREA)

10/100/1000 + inch/foot/yard/mile + acre/hectare + square + verify + audit! Metric + imperial + land + verify + audit — convert + square + audit!

10/100/1000 + inch/foot/mile + audit	Acre/hectare + square + audit	Verify + audit + pride!
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Σ T1: LENGTH — T2: AREA

- T1: mm-cm-m-km + in/ft/yd/mi. Ex: 2.5 km = 2500 m; 12 in = 1 ft.
- T2: square units + acre/hectare. Ex: 5 acres ≈ 20234 m² (4046.86×5).

⚠ FATAL TRAP

Linear-for-area + 'acre = fixed-m' + factor-mix + no-verify = wrong + fail! Square + tables + audit.

★ NORCET HIT

HIT: Cube + cuboid + cylinder next — volume + surface + audit!

► 3.2 m in cm? 2 hectares in m²?

320 cm; 20000 m².

Side³ + 6×side² + space-diagonal + verify + audit! Volume + surface + diagonal + paint + cut + audit — cube + power-three + audit!

$a^3 + 6a^2 + a\sqrt{3}$ + audit

Paint/cut +
verify + audit

Verify + audit
+ pride!

★ CUBE — NEEDS

- Volume/surface/diagonal: $a^3 + 6a^2 + a\sqrt{3}$. Ex: $a5 \rightarrow 125, 150, 5\sqrt{3}$.
- Use: paint + cut + verify + audit. Ex: painted + cut $n = \text{faces} \times n^2 \dots$ (DI link next).

⚠ FATAL TRAP

Area-for-volume + 'diagonal = side' + units-mix + no-verify = wrong + fail! Power + audit.

★ NORCET HIT

HIT: Cuboid + cylinder + cone next — length×width×height + pi + audit!

► Cube 6 volume + surface?

216 + 216.

$L \times W \times H + 2(LW + WH + HL)$ + space-diagonal + verify + audit! Boxes + rooms + tanks + audit — length + surface + audit!

**LWH + 2-sum +
diagonal + audit**

**Boxes/rooms/tanks
+ verify + audit**

**Audit + pride
+ date!**

▲ CUBOID — NEEDS

- Volume/surface/diagonal + verify. Ex: $6 \times 4 \times 3$: V72, S108, $d\sqrt{61}$.
- Use: rooms + tanks + boxes + audit. Ex: paint-4-walls = $2h(L+W)$.

⚠ FATAL TRAP

Cube-formula + 'all sides' + diagonal-miss + units + no-verify = wrong + fail! LWH + audit.

★ NORCET HIT

HIT: Cylinder + cone + sphere next — pi + third + four-thirds!

► $4 \times 3 \times 5$ volume + surface?

60 + 94.

Pi-r²h + 2-pi-r(h+r) + curved + verify + audit! Pipes + tanks + rollers + audit — pi + height + audit!

Pi-r²h + 2-pi-r(h+r)
 + audit

Curved + total +
 verify + audit

Audit + pride
 + date!

◆ CYLINDER — NEEDS

- Volume/CSA/TSA + verify. Ex: r7 h10: V1540, CSA440, TSA748.
- Use: pipes + tanks + hollow + audit. Ex: hollow = outer–inner.

⚠ FATAL TRAP

Circle-area + 'height = slant' + diameter-radius +
 no-verify = wrong + fail! Height + audit.

★ NORCET HIT

HIT: Cone + sphere + hemisphere next — third +
 four-thirds + half!

► r14 h20 (22/7) volume?

22/7×196×20 = 12320.

Third-pi-r²h + pi-r-l + slant + verify + audit! Funnels + tents + hats + audit — third + slant + audit!

Third-pi-r²h +
 pi-r-l + slant

$l^2 = r^2 + h^2 +$
 verify + audit

Audit + pride
 + date!

Σ CONE — NEEDS

- Volume/CSA + slant + verify. Ex: r6 h8: l10, V301.6, CSA188.5.
- Use: funnels + tents + hats + frustum + audit.

⚠ FATAL TRAP

Cylinder-volume + 'height = slant' + third-miss +
 no-verify = wrong + fail! Third + slant + audit.

★ NORCET HIT

HIT: Sphere + hemisphere + conversions next —
 four-thirds + half + units!

► r9 h12 (3.14) volume?

1017.36 ≈ 1017 (third×81×12×3.14).

Four-thirds- πr^3 + 4- πr^2 + diameter + verify + audit! Balls + bubbles + planets + audit —
 four-thirds + four- π + audit!

4/3- πr^3 + 4- πr^2
 + audit

Diameter + verify
 + audit

Audit + pride
 + date!

★ SPHERE — NEEDS

- Volume/surface + verify. Ex: r21: V38808, S5544.
- Use: balls + hemispheres + audit. Ex: hollow = outer-inner.

⚠ FATAL TRAP

Circle-area + '4/3-miss' + diameter-radius + no-verify =
 wrong + fail! Four-thirds + audit.

★ NORCET HIT

HIT: Hemisphere + conversions + melting next — half +
 units + conserve!

► r10.5 (22/7) volume?

$4/3 \times 22/7 \times 1157.6 \approx 4851.$

Two-thirds- $\pi r^3 + 3\pi r^2$ + bowls + domes + audit! Litres + cubic + 1000 + verify + audit — half + convert + audit!

$\frac{2}{3}\pi r^3 + 3\pi r^2$ + audit	Litres/cubic + 1000 + verify	Audit + pride + date!
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▲ HEMI/CAPACITY — NEEDS

- Hemi volume/surface + verify. Ex: r7: V718.7, S462.
- $1 \text{ m}^3 = 1000 \text{ L}$ + verify. Ex: tank $2 \times 1.5 \times 1 \text{ m} = 3000 \text{ L}$.

⚠ FATAL TRAP

Full-sphere + 'litre = m^3 ' + factor-1000-miss + no-verify
 = wrong + fail! Half + 1000 + audit.

★ NORCET HIT

HIT: Melting + DI-tables next — conserve + read +
 audit!

► Hemisphere r14 vol (22/7)? Tank $3 \times 2 \times 1.5$ litres?

5745.3; 9000 L.

MELTING/RECASTING (VOLUME CONSERVED)

Melt + recast + equate + solve + verify + audit! Cubes + spheres + wires + sheets + audit — volume + conserved + audit!

Melt + equate +
solve + verify

Cubes/spheres/wires/
sheets + audit

Conserve + audit
+ pride!

◆ MELT — SOLVES AS

- Equate volumes + solve. Ex: cube $6^3 = 216 \rightarrow$ small $2^3 = 8$ each $\rightarrow 27$ cubes.
- Wires/sheets + audit. Ex: melt sphere $r_6 \rightarrow$ wire r_1 : length = $288\pi/\pi = 288$.

⚠ FATAL TRAP

Surface-equate + 'melt = average' + units-mix + no-verify = wrong + fail! Volumes + audit.

★ NORCET HIT

HIT: DI tables + bars + pies next — read + compute + eliminate!

► Cube 9^3 into 3^3 cubes count?

$729/27 = 27$.

Read + total + average + ratio + percent + max/min + difference + verify + audit! Rows + columns + units + footnotes + audit — read + compute + audit!

Read/total/average/ ratio/percent + audit	Max/min/difference + verify + audit	Footnotes + audit + pride!
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Σ TABLE — DOES

- Read + compute + verify. Ex: total + share + growth + audit.
- Max/min/difference + footnotes + units + audit.

⚠ FATAL TRAP

Eyeball + 'approx fine' + units/footnotes-missed + no-verify = wrong + fail! Read + compute + audit.

★ NORCET HIT

HIT: Bar + pie + line + mixed next — heights + angles + slopes!

► Table + share + growth + total + verify. Steps?

Read + total + ratio + percent + max + audit.

Height + value + compare + total + average + percent + trend + verify + audit! Grouped + stacked + horizontal + scale + audit — heights + values + audit!

Height/value + compare/total + audit	Grouped/stacked + scale + audit	Verify + audit + pride!
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★ BAR — DOES

- Read + value + compare + total + verify + audit.
- Scale + grouped/stacked + legend + audit.

⚠ FATAL TRAP

Eyeball + 'taller = double' + scale-missed + legend-ignored + no-verify = wrong + fail! Scale + legend + audit.

★ NORCET HIT

HIT: Pie + line + mixed + approx next — angles + slopes + eliminate!

► Bar + scale + legend + compare + total. Steps?

Scale + value + compare + percent + audit.

360° + percent + value + sector + compare + verify + audit! Degrees + share + total + legend + audit — angles + shares + audit!

**360° + percent +
 value + sector**

**Compare + total +
 legend + audit**

**Verify + audit
 + pride!**

▲ PIE — DOES

- Degrees/share/total + verify. Ex: 25% of 360 = 90°; total 7200 → 1800.
- Compare + legend + audit. Ex: A:B sectors 120:90 = 4:3.

⚠ FATAL TRAP

Eye-ball-angles + 'biggest = half' + total-missed + no-compute + no-verify = wrong + fail! Degrees + total + audit.

★ NORCET HIT

HIT: Line + mixed + approx next — slopes + combine + eliminate!

► **30% sector + total 5400 value? Ratio 72°:108°?**

1620; 2:3.

Slope + trend + peak/trough + compare + percent + forecast + verify + audit! Time + value + axes + legend + audit — slopes + trends + audit!

**Slope/trend/peak/
trough + audit**

**Compare/percent/
forecast + audit**

**Axes/legend +
verify!**

◆ LINE — DOES

- Read + slope + peak + compare + verify + audit.
- Axes + legend + units + forecast + audit.

⚠ FATAL TRAP

Eye-ball-trend + 'rising = good' + axes-ignored + no-compute + no-verify = wrong + fail! Axes + compute + audit.

★ NORCET HIT

HIT: Mixed + approx + eliminate next — combine + round + options!

► Line + peak + growth% + forecast + verify. Steps?

Axes + values + percent + trend + audit.

Table+bar + pie+line + cross + verify + audit! Round + compatible + options + eliminate + verify + audit — combine + round + eliminate!

Table/bar/pie/line +
 cross + verify

Round/compatible/
 options/eliminate + audit

Combine + audit
 + pride!

Σ MIXED/APPROX — DOES

- Mixed + cross-verify + audit. Ex: table total vs pie share.
- Round + eliminate + verify. Ex: options far → estimate + strike.

⚠ FATAL TRAP

One-source + 'options exact' + no-cross/round + no-eliminate + no-verify = slow + wrong + fail! Cross + round + audit.

★ NORCET HIT

HIT: Full Quant shelf: 73pp + compendium + mocks + revision + audit!

► Options 1.2L/1.5L/1.8L, estimate 1.52L. Pick + why?

1.5L — nearest + verify + eliminate.